

Testing AI Systems

How to test probabilistic systems: response quality, RAG, security, bias, evals and guardrails.

Non-determinism and evals

Generative systems may produce different answers for the same input. Evals use case sets, criteria and metrics to track quality in a repeatable way.

RAG and groundedness

In RAG, test retrieval, source relevance, answer faithfulness to context and behavior when the knowledge base lacks sufficient information.

Security and guardrails

Prompt injection, data leakage, unsafe content, tool abuse and excessive permissions require specific testing plus application and model controls.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

How QA applies it

Start from the behavior or risk to validate, prepare data and environment, execute the technique or tool reproducibly and capture evidence that helps the team understand the result.

Good practices

Prioritize high-impact scenarios, keep data and environments controlled, record important decisions and review the approach periodically to avoid redundant or outdated coverage.

QA tip

In AI, the expected result often needs to be an evaluable criterion rather than an exact sentence.